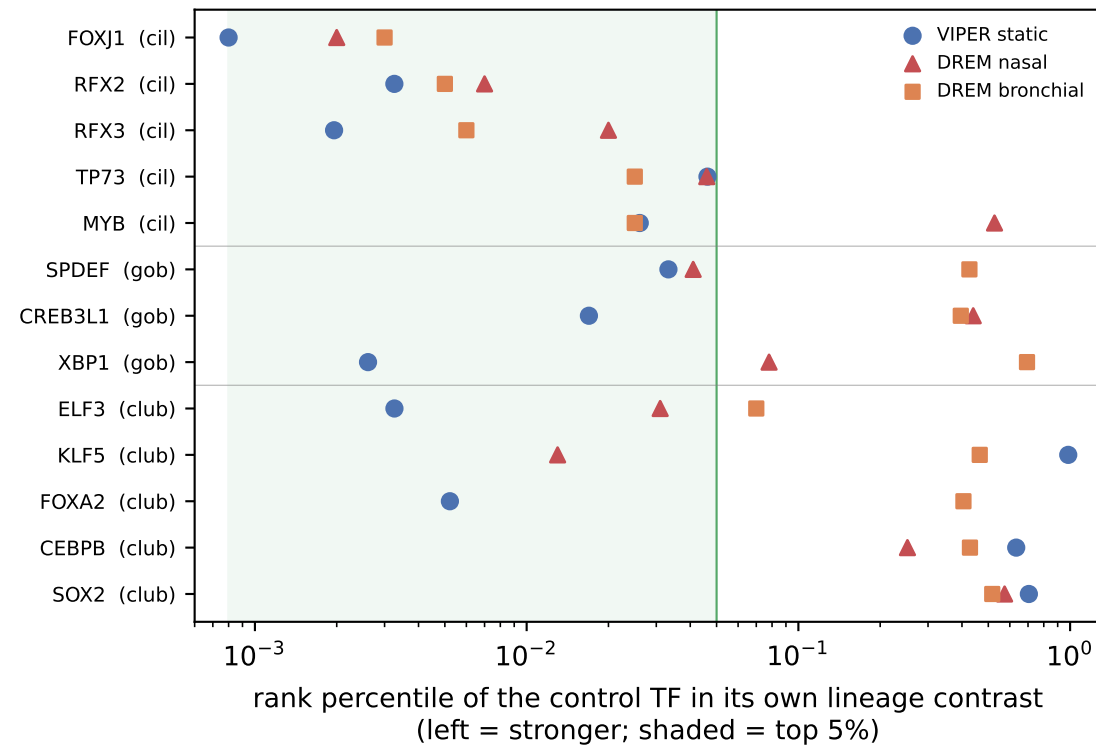
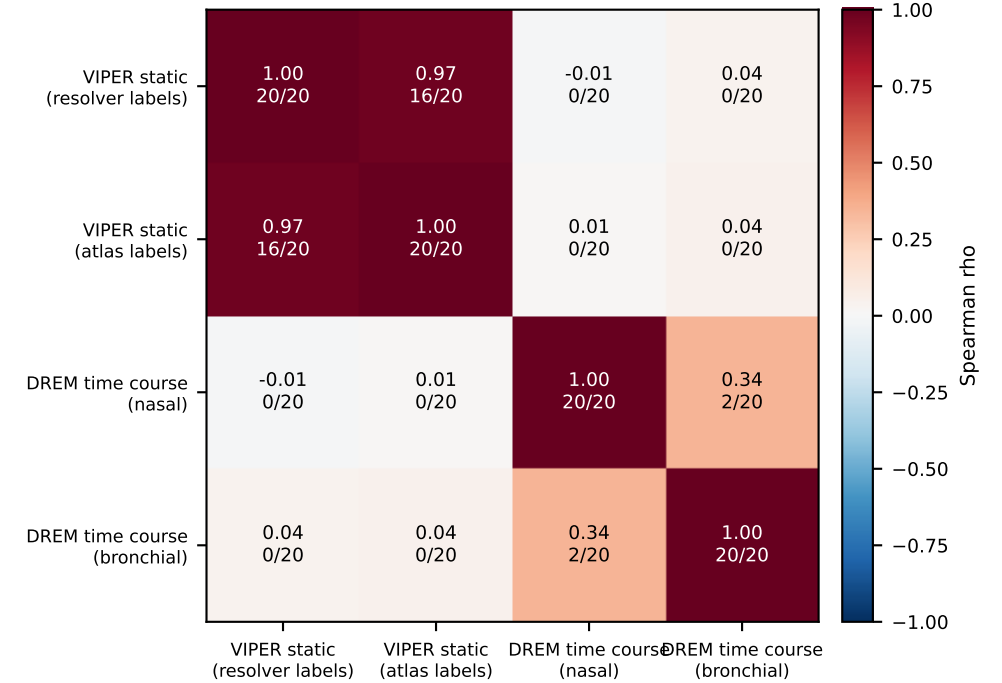


a Controls: ciliated recovered everywhere, goblet only at depth, club priors mid-pack



b Club rankings: Spearman rho and top-20 overlap between methods



c Top-10 candidate club drivers, by method

VIPER static (resolver labels)	VIPER static (atlas labels)	DREM time course (nasal)	DREM time course (bronchial)
1. CREB3L4	1. CREB3L4	1. GRHL3	1. FOXJ1
2. MITF	2. MITF	2. IRF6	2. ZNF474
3. XBP1	3. FOXA1	3. OVOL1	3. CCDC17
4. FOXA1	4. XBP1	4. BARX2	4. RFX3
5. ELF3	5. ELF3	5. SOX15	5. ZNF473
6. MECOM	6. HNF4G	6. ZBTB7C	6. TIGD4
7. CREB3L2	7. STAT1	7. GRHL1	7. ZNF440
8. FOXA2	8. MECOM	8. CASZ1	8. ZBTB7C
9. FOXP4	9. ZNF189	9. PAX9	9. BAZ2A
10. STAT1	10. CREB3L2	10. HES2	10. MGA

no TF is shared by a static and a time-course top-10 list

d The time course does not corroborate the static club TF set

